

3. The table shows the electronic structures of atoms of the elements **A–F**.

**A–F are not the chemical symbols for the elements.**

| Element  | Electronic structure |
|----------|----------------------|
| <b>A</b> | 2                    |
| <b>B</b> | 2,6                  |
| <b>C</b> | 2,8,1                |
| <b>D</b> | 2,8,7                |
| <b>E</b> | 2,8                  |
| <b>F</b> | 2,8,6                |

(a) Which of elements **A–F** is found in Group 6 and Period 3 of the Periodic Table?

Explain your choice, referring to electronic structure.

[2]

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(b) Which **two** of elements **A–F** are chemically inert?

Explain your choice, referring to electronic structure.

[2]

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(c) Element **D** has two isotopes. Isotope **1** has 18 neutrons and isotope **2** has 20 neutrons.

Complete the table by giving the atomic number and mass number of both isotopes. [2]

| Isotope  | Atomic number | Mass number |
|----------|---------------|-------------|
| <b>1</b> |               |             |
| <b>2</b> |               |             |

